

STEM4U — FTC Pre-Inspection Checklist

Get through robot inspection on the first try · BIOBUZZ 2026-27 *Confirm exact numbers on the STEM4U Robot Rules cheat sheet / the current Competition Manual before every event.*

Do this the night before

- ☐ **Batteries charged** — main robot battery + spares, all topped off
- ☐ **Robot fits the sizing cube** in its starting configuration (18" × 18" × 18") — test it in a real box
- ☐ **Team number displayed** on at least two sides, large and readable
- ☐ Robot **powers on cleanly** and drives; Driver Station connects to the Control Hub
- ☐ **Software current** — Robot Controller app and Control Hub OS/firmware updated (REV Hardware Client)
- ☐ Robot Wi-Fi named to your team (not the factory default)

Robot — mechanical

- ☐ Nothing extends outside the sizing cube at the start of a match
- ☐ **No sharp edges, exposed metal, or pinch points** that could hurt people or damage the field
- ☐ All fasteners tight; nothing rattling loose
- ☐ Battery is **securely mounted** (not just resting) and can't fall out
- ☐ Only **one main battery** powering the robot

Robot — electrical

- ☐ **Main power switch** is accessible and clearly reachable
- ☐ Wiring is tidy and strain-relieved — **no exposed conductors**, no wires that can catch
- ☐ Motor/servo counts and controllers are within the allowed limits
- ☐ Only **legal/approved** motors, servos, and electronics
- ☐ Battery connector and main power wiring use the approved connectors

Software & control

- ☐ Robot Controller app version matches the current season
- ☐ Control Hub OS + Hub firmware show "**Up-to-Date**" in REV Hardware Client
- ☐ At least one **OpMode runs** and the robot responds to the gamepad
- ☐ Driver Station device is charged and paired

Bring to inspection

- ☐ The **robot**, on a cart, ready to power on
- ☐ **Charged batteries** (and a spare)
- ☐ Driver Station + gamepads

- ☐ A **copy of your Engineering Portfolio** (judges/inspectors may ask)
- ☐ Basic **tools** for a quick fix (hex keys, screwdrivers, zip ties, tape)

Inspector's mindset: they're checking that the robot is *safe, legal size, and built to the rules* — not whether it's good. Most first-try failures are **size, exposed wiring, loose battery, or team number missing**. Fix those four before you get in line.